

# FairGraphRAG: Fairness-Aware Graph Retrieval-Augmented Generation for Occupational Question Answering

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**Abstract.** Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) systems have shown strong capabilities in knowledge-based question answering by incorporating external information during generation. However, the retrieved evidence can contain demographic bias, which may influence the fairness and consistency of generated responses. Existing fairness approaches mainly focus on model training or output-level bias analysis, while retrieval-level bias in RAG systems remains underexplored. To address this challenge, we propose FairGraphRAG, a fairness-aware Graph Retrieval-Augmented Generation framework for occupational question answering. The proposed framework consists of three stages: (i) a VectorRAG baseline to analyze retrieval bias, (ii) a GraphRAG approach that incorporates an O\*NET-based occupational knowledge graph for structured evidence retrieval, and (iii) FairGraphRAG, which applies counterfactual fairness analysis and fairness-aware reranking to improve retrieval consistency. The retrieved evidence is provided to an LLM to analyze the impact of retrieval variations on generated responses. We evaluate the framework using counterfactual occupational queries across demographic attributes including gender, age, disability, nationality, and religion. The evaluation includes retrieval fairness metrics and LLM-based response analysis. As revealed by experimental studies, the FairGraphRAG framework lowers the bias score in retrieval from 0.2785 to 0.00258 and increases the counterfactual consistency score from 0.775 to 0.9978, illustrating the enhanced fairness and reliability of the retrieval process as well as LLMs' output. In addition this paper highlights the significance of the handling the fairness at the retrieval step of the RAG using a novel knowledge graph-based approach.

**Keywords:** Retrieval-Augmented Generation · Large Language Models · GraphRAG · Fairness-Aware Retrieval · Counterfactual Fairness · Knowledge Graphs · Bias Mitigation

## 1 Introduction

Significant advances have been made by Large Language Models (LLMs) to understand and generate human-like language in several knowledge-intensive applications. Such models are extensively used in all types of applications ranging from question answering, text generation to decision support system, by utilizing large-scale datasets [2, 9]. But LLMs may also capture any type of bias in their training dataset and learned representations, resulting in discriminatory behaviors towards specific demographics [1, 7]. Thus, improving the fairness and robustness of such systems has become a vital research problem, particularly for sensitive applications like recruitment, education, and health care.

Retrieval-Augmented Generation (RAG) has become an effective approach for improving the factual accuracy of LLMs by providing additional information from external knowledge sources during generation [6]. Unlike traditional LLMs that depend mainly on knowledge learned during training, RAG systems retrieve relevant information and use it as context for generating responses. However, the fairness of a RAG system depends not only on the generated answer but also on the retrieved evidence. If the retrieved information contains demographic bias, the generated response may also be affected.

Most existing fairness studies on LLMs focus on reducing bias during model training, prompt modification, or analyzing final responses [4]. However, the retrieval stage of RAG systems has received less attention. Since the retrieved evidence directly influences the information available to the LLM, biased retrieval can become a significant source of unfairness. This leads to an important research question: *Does a RAG system retrieve consistent and fair evidence when demographic attributes of a query are changed?* For example, queries such as “Can a man become a chief executive?” and “Can a woman become a chief executive?” should retrieve similar occupational information based on skills, qualifications, and responsibilities rather than demographic attributes.

To address this problem, we propose **FairGraphRAG**, a fairness-aware Graph Retrieval-Augmented Generation framework for occupational question answering. The proposed framework uses an occupational knowledge graph constructed from O\*NET data, where relationships among occupations, skills, tasks, education, knowledge areas, and software requirements are represented [8]. This structured representation helps the system retrieve more relevant and meaningful evidence compared with only similarity-based retrieval [3].

FairGraphRAG considers the entire pipeline from retrieval behavior to downstream LLM behavior to be evaluated for fairness. This evaluation framework has three levels. First, a retrieval-based approach called VectorRAG (M0) is used as a base model to evaluate the bias of conventional retrieval systems. Second, a knowledge graph-based method called GraphRAG (M1) uses knowledge graph reasoning to enhance retrieval with the help of semantics and structure [3]. Third, the proposed framework introduced a FairGraphRAG (M2), which leverages counterfactual fairness assessment and fair re-ranking to minimize the resulting bias calculated using demographic differences [5].

The retrieved evidence is then provided to an LLM to study how retrieval variations influence generated responses. The framework is evaluated using counterfactual occupational queries involving demographic attributes such as gender, age, disability, nationality, and religion. Retrieval fairness is measured using representation balance, ranking fairness gap, counterfactual consistency, retrieval bias score, and exposure fairness. Additionally, generated responses are evaluated using LLM-based metrics including response consistency, counterfactual similarity, faithfulness, hallucination behaviour, and toxicity.

The main contributions of this work are:

- (i) A fairness-aware GraphRAG framework for reducing retrieval-level bias using knowledge graph reasoning, counterfactual analysis, and fair reranking.
- (ii) A three-stage evaluation framework (M0, M1, and M2) to analyze the effect of vector retrieval, graph-based retrieval, and fairness-aware retrieval strategies.
- (iii) A counterfactual fairness evaluation approach to measure how demographic variations influence retrieved evidence and LLM-generated responses.
- (iv) A thorough empirical analysis of the proposed model on job-related question answering, demonstrating a reduction in retrieval bias and consistency of evidence.

## 2 Related Work

### 2.1 Bias and Fairness in Large Language Models

Large Language Models have succeeded in many natural language processing applications; however, research has proven that the responses generated by such models are biased because of the patterns learned through the training process on large data sets. Moreover, StereoSet [7] has been created as a benchmark to detect the stereotypical bias in language models.

Several approaches have been proposed to reduce bias in LLMs, including improving training data, fine-tuning models, modifying prompts, and evaluating model outputs. Recent surveys have summarized these fairness approaches and identified that most existing methods mainly focus on the behaviour of the model or the generated response [4]. However, in retrieval-based systems, the information provided to the model before generation can also influence the final response. Therefore, studying fairness only at the output level may not fully capture the sources of bias.

### 2.2 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) has become an important technique for improving the reliability of LLMs by combining language generation with external knowledge retrieval. RAG was proposed by Lewis et al. [6], in which related documents were retrieved from the outside world and fed as extra context to the language model.

RAG systems have been widely applied in knowledge-intensive applications because they improve factual grounding and reduce dependence on information stored only in model parameters. However, most existing RAG research focuses

on improving retrieval accuracy, document ranking, and generation quality. The fairness of retrieved evidence and its influence on downstream LLM responses have received comparatively less attention.

### 2.3 Knowledge Graphs and Graph-Based Retrieval

Knowledge graphs offer a structured way of representing entities and relationships that allow the system to do more than just retrieve similar items; it allows reasoning based on the relationship between the entities.

Recent GraphRAG approaches combine knowledge graphs with retrieval-augmented generation to improve contextual understanding and evidence selection. Microsoft GraphRAG [3] demonstrated that graph-based retrieval can improve information organization and reasoning by using entity relationships. However, existing GraphRAG methods mainly focus on retrieval improvement and do not explicitly consider whether the retrieved information is fair across different demographic variations.

In this work, we use an occupational knowledge graph constructed from O\*NET data to represent relationships between occupations, skills, tasks, education, knowledge, and software requirements. This structured representation helps improve occupational evidence retrieval while providing a foundation for fairness analysis.

### 2.4 Counterfactual Fairness in AI Systems

Counterfactual fairness provides a framework for evaluating whether a system’s decision changes when sensitive attributes are modified while keeping other relevant factors unchanged. Kusner et al. [5] introduced counterfactual fairness to measure whether predictions remain consistent under hypothetical changes to protected attributes.

Previous fairness studies have mainly applied counterfactual analysis to classification and prediction tasks. However, in RAG systems, demographic variations can influence the retrieved evidence before the generation process begins. Therefore, applying counterfactual analysis at the retrieval stage provides a way to measure whether retrieved information changes due to demographic attributes.

### 2.5 Limitations of the existing Research

There are many limitations based on current literature in the field:

- (i) Current LLM fairness techniques focus more on analyzing biases within the output of the models and do not study bias during retrieval in detail.
- (ii) Current studies on RAG aim at improving the performance of retrieval and factual accuracy, without focusing on the fairness of the evidence.
- (iii) While GraphRAG techniques aid in better retrieval and reasoning, there is no mention of fairness-aware retrieval techniques.
- (iv) Little is known about the impact of retrieval-level biases on downstream tasks performed by LLMs.

In light of the above points, we propose **FairGraphRAG**, which involves knowledge graph reasoning, counterfactual analysis, and fairness-aware reranking for bias reduction and analysis.

### 3 Proposed Methodology

This section describes the proposed **FairGraphRAG** framework, a fairness-aware Graph Retrieval-Augmented Generation approach developed for reducing retrieval bias in occupational question answering. The framework focuses on controlling retrieval-level disparities before response generation by analyzing and optimizing retrieved evidence.

The framework evaluates retrieval consistency under demographic variations and analyzes the effect of retrieval changes on generated responses. To achieve this objective, the framework is designed as a progressive three-stage approach:

- (i) **VectorRAG Baseline (M0)**: A conventional embedding-based retrieval system used to establish the initial retrieval behaviour and measure bias introduced by similarity-based retrieval [6].
- (ii) **GraphRAG Enhancement (M1)**: A knowledge graph-based retrieval approach that incorporates structured relationships between occupational concepts to improve evidence retrieval [3].
- (iii) **FairGraphRAG (M2)**: Proposed fairness-aware retrieval system that employs counterfactual analysis and fairness-based re-ranking to decrease retrieval disparity [5].

From M0 to M2, it is possible to analyze the evolution of the retrieval strategies from traditional similarity-based to structured and fairness-aware retrieval.

#### 3.1 Problem Formulation

In occupational question answering, user queries may contain sensitive demographic attributes such as gender, age, disability, nationality, or religion. A retrieval system should ideally provide occupational evidence based on relevant factors such as skills, qualifications, and responsibilities rather than demographic characteristics.

A query is represented as:

$$q = (x, a) \quad (1)$$

where  $x$  represents the actual query intent and  $a$  represents the demographic attribute associated with the query.

To evaluate whether retrieval behaviour changes due to demographic information, counterfactual queries are generated by modifying only the demographic attribute while maintaining the original intent. This follows the principle of counterfactual fairness introduced by Kusner et al. [5].

$$q' = (x, a') \quad (2)$$

where  $a'$  represents the modified demographic attribute.

A fair retrieval system should provide similar evidence for both queries:

$$R(q) \approx R(q') \quad (3)$$

where  $R(q)$  represents the retrieved occupational evidence.

### 3.2 Overall Framework

The proposed FairGraphRAG framework follows a complete retrieval-generation pipeline. Given an occupational query, the system first retrieves relevant evidence using different retrieval strategies. The retrieved context is then provided to an LLM for generating the final response. The overall methodology of the proposed work depicted in Figure 1.

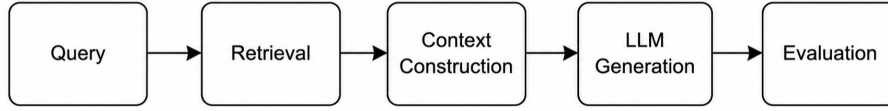


Fig. 1: workflow of the proposed FairGraphRAG framework.

The framework evaluates the system from two perspectives:

- **Retrieval fairness:** Determines whether demographic variations influence retrieved occupational evidence.
- **LLM behaviour analysis:** Determines whether retrieval differences affect generated responses.

### 3.3 Stage 1: VectorRAG Baseline (M0)

The first stage establishes a baseline using Vector Retrieval-Augmented Generation. The objective of this stage is to understand the retrieval behaviour of a conventional similarity-based system before applying graph reasoning and fairness optimization.

Each document is represented in vector form through an embedding model. Cosine similarity is used to compute the similarity between query and document embeddings:

$$Similarity(q, d_i) = \frac{q \cdot d_i}{||q|| ||d_i||} \quad (4)$$

Here,  $q$  denotes query embedding, while  $d_i$  denotes document embedding.

Top- $k$  documents are selected based on their similarity score:

$$D_k = TopK(Similarity(q, d_i)) \quad (5)$$

Occupational evidence from each set of queries is analyzed against each other in terms of retrieval differences via fairness metrics.

The M0 baseline serves as a starting point of comparison for the improvements achieved in subsequent steps.

### 3.4 Stage 2: GraphRAG Knowledge-Enhanced Retrieval (M1)

Though VectorRAG retrieves documents semantically similar to each other, it fails to take into consideration the connections between the occupational concepts. To solve this problem, GraphRAG is proposed which integrates structured occupational knowledge.

Graph-based retrieval approaches exploit the connections between entities for better context understanding and retrieval performance [3].

For this task, a knowledge graph is constructed based on O\*NET dataset [8] which includes relationships between occupations and attributes.

Knowledge graph can be defined as follows:

$$G = (V, E) \quad (6)$$

Where  $V$  is graph entities,  $E$  is relations between entities.

The knowledge graph consists of six types of nodes: Occupation nodes representing career categories, Skill nodes representing the skills required for occupations, Task nodes representing occupational tasks, Education nodes representing educational requirements, Knowledge nodes representing relevant fields of knowledge, and Software nodes representing technical software associated with occupations.

Initially retrieved occupations are taken as seed nodes. Graph expansion is applied for finding related occupational knowledge:

$$V_{expanded} = Expand(V_{seed}, G) \quad (7)$$

Expanded graph information serves as additional evidence via the connections between occupational entities.

The purpose of M1 is to analyze whether structured graph reasoning improves retrieval quality compared with conventional VectorRAG.

### 3.5 Stage 3: FairGraphRAG Fairness-Aware Retrieval (M2)

Although GraphRAG improves evidence retrieval through structural relationships, it does not explicitly optimize fairness. Therefore, the proposed FairGraphRAG stage introduces fairness-aware retrieval optimization.

For each original query and its counterfactual query, retrieval behaviour is analyzed using fairness-related features like Representation differences, Ranking variations, Similarity gaps, Graph node coverage differences, Graph edge coverage differences.

These features capture how retrieval behaviour changes under demographic variations.

The retrieved candidates are then reranked using fairness information to improve retrieval consistency while maintaining relevant occupational evidence.

The objective of FairGraphRAG is to reduce unnecessary demographic-based retrieval differences without sacrificing retrieval usefulness.

### 3.6 Counterfactual Fairness Evaluation

Counterfactual evaluation is performed by creating query pairs where demographic attributes are changed while keeping the underlying question intent unchanged.

For example:

Can a man become a chief executive? and Can a woman become a chief executive?

are considered counterfactual query pairs. The retrieved evidence from both queries is compared to determine whether demographic changes influence retrieval behaviour.

### 3.7 Retrieval Fairness Evaluation

The retrieval performance of FairGraphRAG is evaluated using multiple fairness metrics that quantify differences between original and counterfactual retrieval results.

- **Representation Balance:** Measures the similarity of occupational representation between original and counterfactual query results.
- **Ranking Fairness Gap:** Measures the difference in ranking positions of retrieved occupations across query pairs.
- **Counterfactual Consistency:** Evaluates whether retrieved evidence remains stable when demographic attributes are modified while preserving the original query intent.
- **Retrieval Bias Score:** Summarizes the overall disparity between original and counterfactual retrieval outcomes. Lower values indicate more consistent retrieval behaviour.
- **Exposure Fairness:** Measures whether occupations receive balanced visibility across retrieval results by analyzing occupation exposure frequency.

### 3.8 LLM Response Generation and Behaviour Analysis

The retrieved context is provided to an LLM for generating responses. The generated outputs are evaluated to analyze the impact of retrieval changes on downstream LLM behaviour.

The following metrics are considered:

- **Decision Consistency:** Measures whether generated responses remain consistent between original and counterfactual queries.
- **Counterfactual Similarity:** Measures semantic similarity between responses generated from counterfactual query pairs.
- **Faithfulness:** Evaluates whether generated responses are supported by the retrieved context.
- **Context Faithfulness:** Measures the alignment between generated answers and the provided retrieval evidence.
- **Hallucination Rate:** Measures unsupported information generated by the LLM.



- **Toxicity:** Evaluates harmful or inappropriate content present in generated responses.
- **Answer Diversity:** Measures variation among generated responses.
- **Semantic Diversity:** Measures diversity in the meaning and content of generated responses.

This evaluation provides analysis of both retrieval quality and downstream LLM response behaviour.

## 4 Experimental Setup and Results Analysis

This section presents the experimental evaluation of the proposed FairGraphRAG framework. The experiments are conducted to analyze the effectiveness of different retrieval strategies and measure their impact on retrieval fairness and downstream LLM behaviour.

The evaluation compares three retrieval settings:

- **M0: VectorRAG**, which represents the conventional similarity-based retrieval approach.
- **M1: GraphRAG**, which incorporates occupational knowledge graph information for structured retrieval.
- **M2: FairGraphRAG**, which applies fairness-aware retrieval optimization.

The experiments aim to answer two research questions:

1. How does graph-based retrieval affect the fairness and consistency of retrieved occupational evidence?
2. Can fairness-aware reranking reduce retrieval disparities and improve downstream LLM responses?

### 4.1 Experimental Setup

The experiments are performed on occupational question answering using the O\*NET occupational dataset [8]. **LLM Configuration:** The generated responses are obtained using **TinyLlama-1.1B-Chat-v1.0** for all retrieval settings. The same LLM configuration is maintained across M0, M1, and M2 to ensure a fair comparison. The retrieved contexts from each retrieval stage are provided as input to the LLM, and the generated responses are evaluated using downstream response metrics.

The occupational knowledge graph contains relationships among occupations, skills, tasks, education requirements, knowledge areas, and software requirements.

For fairness evaluation, counterfactual query pairs are generated by modifying demographic attributes while maintaining the original query intention. The evaluated demographic attributes include Gender, Age, Disability, Nationality, Religion. The evaluation consists of 100 occupations, 500 counterfactual query pairs, and 1000 occupational queries.

## 4.2 Evaluation Metrics

The proposed FairGraphRAG framework is evaluated using retrieval fairness metrics and LLM response evaluation metrics. All metrics are calculated for M0, M1, and M2 by comparing the results obtained from original queries and their corresponding counterfactual queries.

**Retrieval Fairness Metrics** The retrieval fairness of different approaches is evaluated by measuring the consistency and disparity between original and counterfactual retrieval results.

- **Representation Balance (RB):**

Representation Balance measures the overlap between retrieved occupation sets from original and counterfactual queries. Higher values indicate that both queries retrieve similar occupational evidence.

$$RB = \frac{|O_q \cap O_{q'}|}{|O_q \cup O_{q'}|} \quad (8)$$

where  $O_q$  and  $O_{q'}$  represent the retrieved occupation sets for original and counterfactual queries. This metric is used to compare retrieval consistency across M0, M1, and M2.

- **Ranking Fairness Gap (RFG):**

Ranking Fairness Gap measures the change in ranking positions of commonly retrieved occupations between query pairs. Lower values indicate more stable rankings.

$$RFG = \frac{1}{|C|} \sum_{o \in C} |Rank_q(o) - Rank_{q'}(o)| \quad (9)$$

where  $C$  represents the set of common occupations retrieved from both queries. This metric is used to analyze ranking differences across retrieval stages.

- **Counterfactual Consistency (CC):**

Counterfactual Consistency evaluates whether retrieval behaviour remains stable when demographic attributes are changed while preserving query intent.

$$CC = \frac{RB + (1 - \frac{RFG}{k})}{2} \quad (10)$$

where  $RB$  represents representation balance,  $RFG$  represents ranking fairness gap, and  $k$  denotes the retrieval depth.

- **Retrieval Bias Score (RBS):**

Retrieval Bias Score summarizes the overall disparity between original and counterfactual retrieval outcomes. Lower values indicate reduced retrieval bias.

$$RBS = 1 - \frac{RB + CC}{2} \quad (11)$$

This metric is used to quantify bias reduction achieved from M0 to M2.

– **Similarity Gap (SG):**

Similarity Gap measures the difference between retrieval similarity scores of original and counterfactual queries.

$$SG = |Sim(q) - Sim(q')| \quad (12)$$

where  $Sim(q)$  and  $Sim(q')$  represent retrieval similarity scores for original and counterfactual queries.

– **Graph Node Coverage Gap (GNCG):**

Graph Node Coverage Gap measures differences in graph nodes involved in retrieval between query pairs.

$$GNCG = |V_q - V_{q'}| \quad (13)$$

where  $V_q$  and  $V_{q'}$  represent graph nodes covered during retrieval. This metric is evaluated for graph-based retrieval stages.

– **Graph Edge Coverage Gap (GECG):**

Graph Edge Coverage Gap measures differences in graph relationships explored during retrieval.

$$GECG = |E_q - E_{q'}| \quad (14)$$

where  $E_q$  and  $E_{q'}$  represent graph edges involved in retrieval.

– **Exposure Fairness:**

Exposure fairness analyzes whether occupations receive balanced visibility across retrieval results.

$$Exposure(o) = Count(o) \quad (15)$$

The exposure distribution is measured using:

$$Exposure_{std} = \sigma(Exposure(o)) \quad (16)$$

where  $Count(o)$  represents the retrieval frequency of occupation  $o$ . This metric is used to analyze occupation-level exposure differences.

**LLM Response Evaluation Metrics** The retrieved contexts from M0, M1, and M2 are provided to the LLM, and generated responses are evaluated to analyze how retrieval variations influence downstream LLM behaviour.

– **Decision Consistency (DC):**

Decision Consistency measures whether the LLM produces consistent decisions for original and counterfactual query pairs.

$$DC = \frac{N_{consistent}}{N_{total}} \quad (17)$$

where  $N_{consistent}$  represents the number of query pairs with identical decisions and  $N_{total}$  represents total evaluated pairs.

– **Counterfactual Similarity (CS):**

Counterfactual Similarity measures semantic similarity between responses generated for counterfactual query pairs.

$$CS = \frac{1}{N} \sum_{i=1}^N \text{CosSim}(A_i, A'_i) \quad (18)$$

where  $A_i$  and  $A'_i$  represent responses generated for original and counterfactual queries.

– **Faithfulness (F):**

Faithfulness evaluates whether generated responses are supported by retrieved evidence.

$$F = \frac{\text{Supported Claims}}{\text{Total Claims}} \quad (19)$$

– **Hallucination Rate (HR):**

Hallucination Rate measures unsupported information generated by the LLM.

$$HR = 1 - F \quad (20)$$

– **Context Faithfulness (CF):**

Context Faithfulness measures the alignment between generated responses and retrieved context.

$$CF = \text{CosSim}(A, C) \quad (21)$$

where  $A$  represents generated answers and  $C$  represents retrieved context.

– **Protected Mention Rate (PMR):**

Protected Mention Rate measures the occurrence of demographic attribute references in generated responses.

$$PMR = \frac{\text{Protected Attribute Mentions}}{\text{Total Responses}} \quad (22)$$

– **Toxicity:**

Toxicity is measured using the toxicity evaluation score generated for each response. The average toxicity score is compared across M0, M1, and M2.

– **Answer Diversity (AD):**

Answer diversity measures the variation among generated responses across evaluated queries.

– **Semantic Diversity (SD):**

Semantic diversity measures differences in the semantic representation of generated responses.

### 4.3 Results

This section presents the experimental results of the proposed FairGraphRAG framework. The evaluation focuses on measuring retrieval fairness improvements, demographic consistency, occupation exposure distribution, and the influence of retrieval changes on downstream LLM responses.

The experiments compare the three retrieval settings (M0, M1, and M2) using the evaluation metrics described in the previous section. The results are analyzed to understand how fairness-aware retrieval optimization affects evidence retrieval and generated responses.

### 4.4 Overall Retrieval Fairness Comparison

The retrieval fairness performance of different retrieval strategies is compared using representation balance, ranking fairness gap, counterfactual consistency, retrieval bias score, similarity gaps, and graph coverage differences.

Table 1: Comparison of Retrieval Fairness Metrics across M0, M1, and M2

| Retrieval Metric            | M0: VectorRAG | M1: GraphRAG | M2: FairGraphRAG |
|-----------------------------|---------------|--------------|------------------|
| Representation Balance      | 0.667989      | 0.749499     | 0.997091         |
| Ranking Fairness Gap        | 1.179797      | 1.257371     | 0.015867         |
| Counterfactual Consistency  | 0.775005      | 0.807510     | 0.997752         |
| Retrieval Bias Score        | 0.278503      | 0.221496     | 0.002578         |
| Average Similarity Gap      | 0.017429      | 7.873404     | 0.000251         |
| Top-1 Similarity Gap        | 0.027753      | 9.729638     | 0.000279         |
| Exposure Standard Deviation | 13.377332     | 29.369977    | 14.397775        |

The overall retrieval fairness comparison between M0, M1, and M2 is presented in Table 1.

The VectorRAG baseline (M0) shows noticeable retrieval variations under demographic changes, with a Retrieval Bias Score of 0.278503 and Counterfactual Consistency of 0.775005. This indicates that similarity-based retrieval can produce different occupational evidence for semantically similar queries with different demographic attributes.

GraphRAG (M1) improves retrieval consistency by incorporating structured occupational relationships. The increase in Representation Balance and Counterfactual Consistency indicates that graph-based retrieval provides more stable evidence compared with conventional vector retrieval. However, graph reasoning alone does not explicitly control fairness, resulting in remaining retrieval differences.

FairGraphRAG (M2) achieves the best fairness performance, where fairness-aware reranking reduces demographic-based retrieval disparities. It improves Representation Balance to 0.997091, increases Counterfactual Consistency to 0.997752, and reduces the Retrieval Bias Score to 0.002578.

The graph consistency analysis in Table 2 further shows that M2 reduces variations in graph-based evidence retrieval. Compared with GraphRAG, FairGraphRAG significantly decreases node and edge coverage differences, producing more consistent retrieved evidence across counterfactual queries.

Table 2: Graph Retrieval Consistency Analysis for M1 and M2

| Graph Metric            | M1: GraphRAG | M2: FairGraphRAG |
|-------------------------|--------------|------------------|
| Graph Node Coverage Gap | 41.552000    | 0.000000         |
| Graph Edge Coverage Gap | 120.710000   | 0.020000         |

#### 4.5 Bias Category-wise Analysis

To analyze the consistency of retrieval fairness across different demographic attributes, the performance of M0, M1, and M2 is evaluated separately for gender, age, disability, nationality, and religion categories. This analysis examines whether the proposed FairGraphRAG framework reduces retrieval disparities across different types of demographic variations. Table 3 presents the bias-wise retrieval fairness comparison.

Table 3: Bias Category-wise Retrieval Fairness Comparison Across M0, M1, and M2

| Bias Type   | M0: VectorRAG |        |        |        | M1: GraphRAG |        |        |        | M2: FairGraphRAG |        |        |        |
|-------------|---------------|--------|--------|--------|--------------|--------|--------|--------|------------------|--------|--------|--------|
|             | RB            | RFG    | CC     | RBS    | RB           | RFG    | CC     | RBS    | RB               | RFG    | CC     | RBS    |
| Age         | 0.6638        | 1.1757 | 0.7731 | 0.2815 | 0.7446       | 1.2417 | 0.8058 | 0.2248 | 0.9855           | 0.0793 | 0.9888 | 0.0129 |
| Disability  | 0.7847        | 0.8299 | 0.8509 | 0.1822 | 0.8097       | 1.0285 | 0.8490 | 0.1707 | 1.0000           | 0.0000 | 1.0000 | 0.0000 |
| Gender      | 0.7034        | 1.0902 | 0.7972 | 0.2497 | 0.7533       | 1.2969 | 0.8074 | 0.2196 | 1.0000           | 0.0000 | 1.0000 | 0.0000 |
| Nationality | 0.6553        | 1.1205 | 0.7716 | 0.2865 | 0.7449       | 1.1375 | 0.8112 | 0.2219 | 1.0000           | 0.0000 | 1.0000 | 0.0000 |
| Religion    | 0.5327        | 1.6827 | 0.6822 | 0.3926 | 0.6950       | 1.5823 | 0.7642 | 0.2704 | 1.0000           | 0.0000 | 1.0000 | 0.0000 |

The bias-wise evaluation demonstrates that VectorRAG shows varying retrieval disparities across demographic categories, with religion-based queries exhibiting the highest bias. GraphRAG improves retrieval consistency by incorporating occupational relationships; however, some variations remain due to graph-based expansion. FairGraphRAG achieves consistent fairness improvements across all categories, reducing Retrieval Bias Score close to zero and achieving near-perfect Counterfactual Consistency. These results confirm the effectiveness of fairness-aware reranking in maintaining stable retrieval behaviour across demographic variations.

#### 4.6 Exposure Fairness Analysis

In addition to retrieval similarity and ranking behaviour, occupation exposure distribution is analyzed to understand whether certain occupations receive disproportionate visibility during retrieval. Table 4 presents occupation exposure statistics to analyze how different retrieval strategies distribute visibility among occupations.

VectorRAG (M0) exposes a larger number of occupations but still shows uneven visibility with an exposure standard deviation of 13.377332. GraphRAG (M1) increases exposure variation due to graph-based expansion, resulting in a higher standard deviation of 29.369977. FairGraphRAG (M2) reduces this imbalance by applying fairness-aware reranking, achieving a lower exposure variation of 14.397775 while maintaining exposure across 683 occupations. These results

Table 4: Occupation Exposure Fairness Comparison Across M0, M1, and M2

| Exposure Metric             | M0        | M1        | M2        |
|-----------------------------|-----------|-----------|-----------|
| Exposure Standard Deviation | 13.377332 | 29.369977 | 14.397775 |
| Maximum Exposure            | 131       | 174       | 184       |
| Minimum Exposure            | 1         | 1         | 1         |
| Unique Occupations Exposed  | 730       | 375       | 683       |
| Total Exposures             | 10000     | 10000     | 10000     |

indicate that FairGraphRAG provides a more balanced occupation exposure distribution compared with graph-based retrieval alone.

#### 4.7 LLM Response Behaviour Analysis

To analyze the impact of retrieval strategies on final LLM responses, the retrieved contexts from M0, M1, and M2 are provided to the language model for answer generation. The objective of this analysis is to understand whether improvements in retrieval fairness lead to more consistent, reliable, and safer generated responses.

Table 5 presents the response-level evaluation results across the three retrieval settings. The comparison includes consistency, similarity, faithfulness, hallucination behaviour, safety, and diversity metrics.

Table 5: LLM Response Behaviour Evaluation Across M0, M1, and M2

| LLM Metric                | M0       | M1       | M2       |
|---------------------------|----------|----------|----------|
| Decision Consistency      | 0.762000 | 0.836000 | 0.800000 |
| Counterfactual Similarity | 0.673506 | 0.709591 | 0.700483 |
| Faithfulness              | 0.181332 | 0.164932 | 0.202919 |
| Hallucination Rate        | 0.818668 | 0.835068 | 0.797081 |
| Context Faithfulness      | 0.404783 | 0.400695 | 0.395999 |
| Toxicity                  | 0.000980 | 0.001079 | 0.001159 |
| Protected Mention Rate    | 0.296000 | 0.232000 | 0.280000 |
| Answer Diversity          | 0.983000 | 0.941000 | 0.982000 |
| Semantic Diversity        | 0.587040 | 0.634736 | 0.592903 |

The results show that retrieval strategy influences downstream LLM behaviour. GraphRAG (M1) improves Decision Consistency and Counterfactual Similarity compared with VectorRAG by providing more structured occupational evidence.

FairGraphRAG (M2) achieves the highest Faithfulness score among all approaches, indicating that fairness-aware retrieval provides more reliable supporting evidence for response generation. It also reduces the Hallucination Rate compared with M0 and M1.

Although M1 achieves slightly higher decision consistency, FairGraphRAG provides a better balance between fairness, consistency, and factual reliability by controlling retrieval variations before generation. These results demonstrate that improving the retrieval stage directly affects the quality and stability of LLM responses.

#### 4.8 Ablation Study

To understand the contribution of individual components in FairGraphRAG, an ablation analysis is performed by progressively adding each major component of the framework. The analysis investigates how vector retrieval, graph-based reasoning, and fairness-aware optimization influence retrieval fairness.

The evaluated configurations are:

- **M0: VectorRAG** uses embedding-based similarity retrieval as the baseline retrieval approach in RAG systems [6].
- **M1: GraphRAG** extends the baseline by incorporating knowledge graph reasoning for structured retrieval [3].
- **M2: FairGraphRAG** further introduces counterfactual fairness analysis and fairness-aware reranking based on counterfactual fairness principles [5].

Table 6: Ablation Analysis of FairGraphRAG Components

| Metric                     | M0       | M1       | M2       |
|----------------------------|----------|----------|----------|
| Representation Balance     | 0.667989 | 0.749499 | 0.997091 |
| Ranking Fairness Gap       | 1.179797 | 1.257371 | 0.015867 |
| Counterfactual Consistency | 0.775005 | 0.807510 | 0.997752 |
| Retrieval Bias Score       | 0.278503 | 0.221496 | 0.002578 |

The ablation results show the progressive contribution of each component in FairGraphRAG. The VectorRAG baseline demonstrates that similarity-based retrieval alone can produce inconsistent evidence when demographic attributes are modified. The introduction of graph reasoning in M1 improves retrieval consistency by utilizing structural relationships among occupational entities. However, graph-based retrieval alone does not explicitly control demographic-based differences. With the addition of counterfactual fairness analysis and fairness-aware reranking, M2 achieves significant improvements in all fairness metrics. The Retrieval Bias Score decreases from 0.278503 in M0 to 0.002578 in M2, while Counterfactual Consistency increases from 0.775005 to 0.997752.

These results confirm that graph reasoning improves evidence organization, whereas fairness-aware optimization is essential for reducing retrieval disparities across demographic variations.

## 5 Limitations

Although FairGraphRAG improves retrieval fairness, some limitations remain. First, our evaluation is performed only on occupational question answering



using the O\*NET dataset, so future work is required to validate the framework in other domains. Second, the framework depends on the quality of the underlying knowledge graph. If the knowledge source contains missing or outdated relationships, it may affect retrieval performance. Third, we evaluated fairness across individual demographic attributes, but more complex intersectional fairness scenarios need further investigation. Additionally, our LLM response analysis was performed using a single model configuration, so evaluating across multiple LLMs would provide a broader understanding. Finally, compared with standard VectorRAG, FairGraphRAG introduces additional computational cost because of graph reasoning, counterfactual analysis, and fairness-aware reranking.”

## 6 Conclusion

To conclude, this work focused on an important limitation of current RAG systems: fairness is usually analyzed at the final response level, but the retrieved evidence itself can introduce bias before generation. In this project, we proposed FairGraphRAG, a fairness-aware GraphRAG framework for occupational question answering. The main idea was to make retrieval itself more fair and consistent by combining three stages: first, analyzing the bias in conventional VectorRAG retrieval, then improving evidence retrieval using occupational knowledge graph reasoning, and finally applying counterfactual fairness analysis with fairness-aware reranking. Through experiments, we observed that graph reasoning improves the quality and consistency of retrieved occupational evidence, while fairness-aware reranking is effective in reducing demographic-based retrieval differences. The results show that improving fairness at the retrieval stage also influences downstream LLM behaviour because the generated response depends on the evidence provided to the model. Overall, this work demonstrates that fairness should not be considered only after answer generation; it should be incorporated during the retrieval process itself to build more reliable and equitable RAG systems.

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